



Reference Architecture for Project, Program and Portfolio Governance

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Abstract. This paper presents a reference architecture on projects, programs and portfolios (PPP) governance models. Usually, organizations have projects organized into programs that in turn are organized into portfolios. Within organizations it is necessary to establish project, program and portfolio management policies, so that all the expected benefits of projects are achieved and the previously imposed constraints (time, costs, etc.) are met. It is therefore also important for organizations to have a governance model for projects, programs and portfolios. A governance model entails how decisions are made and how actions are carried out, considering organizational values. The required roles, responsibilities, and performance criteria should be an integral part of the governance model for projects, programs, and portfolios. However, given the permeability of projects, programs, and portfolios to several factors (e.g. regulatory factors and the project's project management maturity), several roles can be considered about governance models, thus promoting the existence of different project, program and portfolio governance models. This research proposes a reference architecture that allows the comparison of governance models to verify deviations and detect which model best suits the context of an organization.

Keywords: Projects · Programs · Portfolios · Project management · Governance · Reference architecture · Enterprise Architecture · Competences · Layers · Roles

1 Introduction

Nowadays, organizations live immersed in a constantly changing and extremely competitive market. This change and competitiveness influence the definition of the organizations' strategy. An organization's strategy, in turn, contributes to the continuous improvement of the business [20], supporting the development of organizations [15]. Projects must bring value to organizations as well as the expected return on investment. For this to happen, it is necessary to carry out a correct management of projects, programs and portfolios (PPP) and to follow a correct PPP governance model. Portfolio consists of structured knowledge about the programs and projects underway in the organization, views of costs, terms, scope, among others, are characteristics to which the portfolio has

access [17]. Programs aim to organize projects according to their dominant scope [16]. A project is a temporary effort made to create a product, service or result [15].

For the success of projects, programs and portfolios it is necessary a proper project, program and portfolio (PPP) management. “Project management performance explained 44,9% of the variance in project success”, [21] and follow a correct PPP governance model, “approximately 6.3% of the variation of project success correlates with the stakeholder-orientation of the governance structure” [19].

The term governance in this investigation is considered as “a formal or informal model that determines how decisions are made and how actions are carried out, in the perspective of maintaining organizational values in face of changes, whether these are caused by problems, changes in actors or changes in environments”, OECD definition [26].

“An effective PPP governance model ensures that the project portfolio is aligned with the organization’s objectives (...) and aligns the interests of the directors, project team and other stakeholders” [2], “the governance of a project must influence the selection of the project management methodology and the way it evolves” [31].

However, for each company and its context, defining the best PPP governance model is really a problem.

This research is a extension of the paper “*Project, Program, Portfolio Governance Model Reference Architecture in the Classic Approach to Project Management*” [6] that leads with a governance model related to PPP management systems. We take a PPP governance model as regulations of PPP management, it can include policies, procedures, roles, responsibilities and other things [18]. Considering the probably of the environment affect projects, programs and portfolios, different roles can be considered when referring to PPP governance models.

To allow PPP governance help PPP management is essential that the roles associated with the PPP governance model has the competencies associated with them.

This research proposes a reference architecture that can be used to find comparisons between PPP governance models and help organizations find the best PPP governance model that best suits their needs.

Next section presents this research related work. In Sect. 3 the governance model reference architecture is proposed. The demonstration and evaluation are performed in Sects. 4 and 5, respectively. Section 6 presents the conclusions and the future work.

2 Related Work

This section introduces the main areas relevant for the development of this research.

2.1 PPP Governance

The concept of governance is understood in this research as “a formal or informal model that determines how decisions are made and how actions are taken, with a view to maintaining organizational values in face of change, whether caused by problems, changes in actors or changing environments”- OECD [13].

An effective PPP governance model “ensures that the project portfolio is aligned with the organization’s objectives (...) and will align the interests of directors, project team and other stakeholders” [2]. The governance model gains importance regarding the alignment between projects and organizational objectives and the delivery of project results in an efficient and sustainable manner.

“Roles, responsibilities and performance criteria should be clearly defined in the PPP governance model” [2]. “Role” in an organization is defined within this research as a “function or something someone has” [28].

It is therefore essential that PPP governance is aligned with organizational governance, respecting its principles, decisions and processes [1].

Organizational Governance is a model that includes, “rules, processes, norms, relationships, systems and procedures” [1]. This model allows to steer an organization [18] by influencing how the organization’s objectives are defined and achieved, how risk is monitored and evaluated, and hoe performance is optimized [1].

In the governance model present in ISO 21505 [18], four key roles and their responsibilities related to PPP governance and two related Organizational Governance are presented. The roles are identified in Table 1.

Table 1. Roles related to PPP governance and organizational governance [6].

PPP governance	Organizational governance
PPP governing body	Governing body
Project governing body	Organizational governing body
Program governing body	
Portfolio governing body	

2.2 PPP Management

Project management gains importance in organizations, enabling strategy implementation, business transformation and the development and continuous improvement of new products [21].

The “Rethinking Project Management (RPM) [34] research line contributed to “complement the classic approach to project management represented in PMBOK [1]” [29] and similarly in ISO 21500 [15], 21503 [16], 21504 [17] in PM2 [8] and APMBok [2].

The line of research proposed by Sauer and Horner [35], aim to “highlight the criticisms of the project management knowledge base, (...) provide a set of possible research directions that can be followed in order to enrich and increase the project management knowledge base” [35]. The knowledge base that was criticized at the time is still used in many organizations and is considered in this research as a classical approach to project management, which is characterized by:

- Project and Project Management Lifecycles;
- Projects as Instrumental Processes;
- Focus on Product Creation;
- Restricted Project Conceptualization;
- Practitioners as Trained Technicians;

Knowing that there are several PPP management models, each with their own defined set of roles, it was decided to use as reference in this research the model presented in ISO's, namely ISO 21500 [15], for project management related roles, ISO 21503 [16], for program management related roles and ISO 21504 [17] for portfolio management roles, because the set of roles presented is referenced in different methodologies such as [1, 2], and is summarized in Table 2.

To help organizations compare their practices with the best ones, project management maturity models, “enable an organization to evaluate and compare its own practices with best practices and purpose improvements. A maturity model is a framework that describes the optimal progression using various levels until the desired improvement is achieved” [23]. Examples of project management maturity models are: i) the Organizational Project Management Maturity Model (OPM3) [9]; ii) the Capability Maturity Model Integration (CMMI) [30]; and iii) the IT Score for Program & Portfolio Management [12, 24].

Table 2. Roles related to PPP management [6].

ISO 21500 (Projects)	ISO 21503 (Programs)	ISO 21504 (Portfolio)
Project sponsor	Program sponsor	Portfolio manager
Project manager	Program manager	Portfolio management team
Project management team	Program management team	
Project team		

2.3 Organizational Layers

“Structuring an organization is important because it defines how tasks are divided and how organizational resources will be implemented” [7]. The organizational structure is defined as, (1) “the set of tasks assigned to individuals and departments”; (2) “dependency relationships within organizations, including lines of authority, decision-making responsibility, number of hierarchical levels, and the number of people that managers are responsible for controlling”; (3) and finally the “systems used by the organization to ensure coordination among departmental officials” [3].

For PPP governance and management, the PM2 methodology [8] creates a layered structure where it is possible to relate each role to its layer, taking into account the characteristics of both.

These layers are:

- Business Governing Layer – “Determines the vision and strategy for the entire organization. It consists of one or more management committees that operate at the level of the directors. This is where priorities are set, investment decisions are made and resources are allocated”.
- Steering Layer – “Provides guidance and general project guidance, keeping the project focused on its goals”.

- Directing Layer – “Sponsors the project and has its Business Case. Mobilize necessary resources and monitor project performance to achieve project objectives”.
- Managing Layer – “It focuses on the daily operations of the project, organizing, monitoring and controlling the work to produce the desired deliverables and implement them in the organization. Managing Layer members report to Directing Layer”.
- Performing Layer – “Performs the project work, producing deliveries and implementing them in the organization. Members belonging to the Performing Layer report to Managing Layer”.

Other methodologies indirectly also present a layered structure. For example, the ISO suggests for governance a layered structure where there is a main Organizational Governance layer, which encompasses PPP Governance which in turn encompasses portfolios, programs and projects. For PPP management the ISO suggests a similar structure for projects, programs and portfolios, with a sponsorship layer, managing layer and developing layer (Fig. 1).



Fig. 1. Layered structure related to the PPP governance model present in ISO 21505 [5].

2.4 PPP Management Competences

The training of people involved in projects must be a concern of project managers as well as other aspects of human resource management practices related to projects that until then had been neglected, including the delegation of roles and responsibilities, the identification of informal communication networks and their risks and constant monitoring of the requirements required for the project and the capabilities of human resources [27].

IPMA ICB is a “global standard that defines the competences required of individuals working in the field of project, program and portfolio management” [14].

Regarding projects, programs and portfolios, the competencies in IPMA ICB are divided into three domains: i) “People competences”, the personal or interpersonal skills required to successfully participate or lead a project, program or portfolio [14]; ii) “Practice competences”, specific methods, tools and techniques used in projects, programs or portfolios to make them successful [14]; and iii) “Perspective Competences”, specific methods, tools and techniques by which individuals interact with their surroundings and lead organizations to initiate and support projects, programs or portfolios [14] (Fig. 2).

Perspective	Strategy	Practice	Design
	Governance, structures and processes		Requirements, objectives and benefits
	Compliance, standards and regulations		Scope
	Power and interest		Time
	Culture and values		Organisation and information
People	Self-reflection and self-management		Quality
	Personal integrity and reliability		Finance
	Personal communication		Resources
	Relationships and engagement		Procurement and partnership
	Leadership		Plan and control
	Teamwork		Risk and opportunities
	Conflict and crisis		Stakeholders
	Resourcefulness		Change and transformation
	Negotiation		Select and balance
	Result orientation		

Fig. 2. The 3 areas of competence associated with project, program and portfolio management.

2.5 Reference Architecture of the PPP Governance Model

“A reference architecture captures the essence of other existing architectures, as well as their vision and evolving needs to provide guidance in developing new architectures” [4]. Reference architectures “allow to be effective, (...) discuss future modifications and extensions” [4], however this type of architecture must “be based on proven concepts” [4] and is considered a “standard method (...)” in line with best practice in the area” [4].

There are already investigations aimed at building a reference architecture of the PPP governance model, one of which is [33] that propose four key elements for improving project performance and therefore create value for organizations. These four elements are: (1) Portfolio Management; (2) Project Sponsor; (3) Project Management Office (PMO); (4) Project and Program Support. Also the research [25], with the aim of creating a framework that is useful for project governance, clarifies the various dimensions that the governance concept can acquire: 1) sovereignty, a popular dimension of governance often found in the general management literature, which describes the levels of external autonomy and internal control granted to projects by the governance system; 2) governance mechanisms, predominantly control oriented; and 3) the number of governance models as an indicator of the complexity of project governance implementation.

2.6 Enterprise Architecture

“An organization-level architecture is called Enterprise Architecture” [11], which by definition, according to Lankhorst, consists of “a coherent set of principles, methods and models used in the design and realization of the organizational structure of a company, business processes, information systems and infrastructure” [22]. ArchiMate [32], is a language that provides a uniform graphical representation of Enterprise Architecture [22].

This research uses the Implementation and Migration Elements as well as the Business Actor, Business Role, Business Process and Business Function, business elements and Capability, strategy element, of ArchiMate [32].

3 Solution Proposal

Due to the increasing number of existing PPP governance models and the different contexts in which they can be applied, there is a need to create a reference architecture to the PPP governance.

This architecture is expected to provide a baseline to identify the differences between different PPP governance models in a classical approach to project management.

The reference architecture is divided into layers, where one can have access to the roles and competencies associated with each layer. The reference architecture is developed through 7 steps, next described.

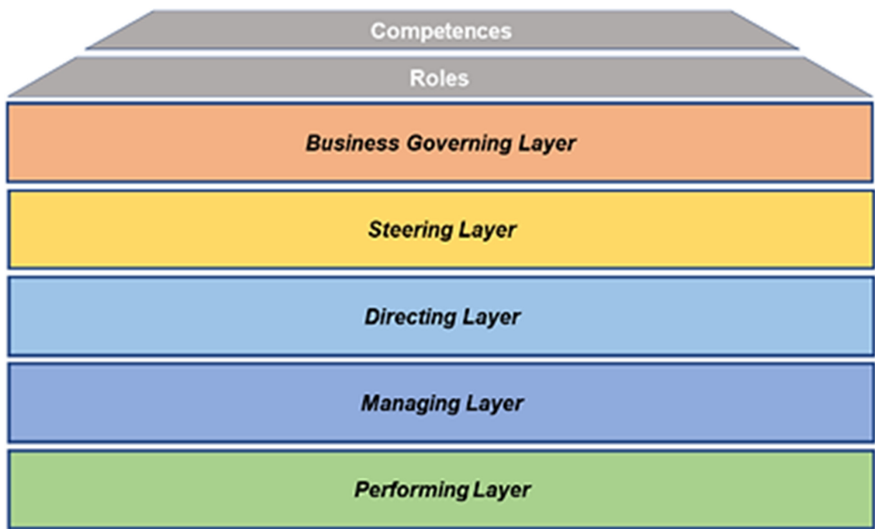


Fig. 3. Reference architecture layers [6].

3.1 Define Architecture Layers

The decision to divide the architecture into layers arises due to the fact that a structured architecture in layers “defines how tasks are divided and how the organization’s resources will be implemented” [27], thus allowing a better understanding of the organization’s command line as well as the specialization of the work.

In order to choose layers for the reference architecture, the different existing layers are analyzed. The PM2 methodology [8] introduces the division of the project management organization in layers that allow a clear relationship between the definition of functions and the roles associated with them and the level that they integrate within organizational

structures and processes. More specifically the PM2 layers, allow a more organizational-related approach that integrates the Business Governing Layer while the other layers are more directed to the project domain, programs and portfolios – see Fig. 3.

3.2 Define Architecture Roles

Organizational governance, as the system by which organizations are directed and controlled [13], has in the roles associated with PPP governance and management processes the core activities carried out under the organizational governance system. These organizational roles, defined as “the roles played by individuals or groups in a project” [2], program or portfolio, are considered fundamental and essential in creating a reference architecture of the PPP governance model. An analysis of existing governance and management models is performed (Fig. 4).

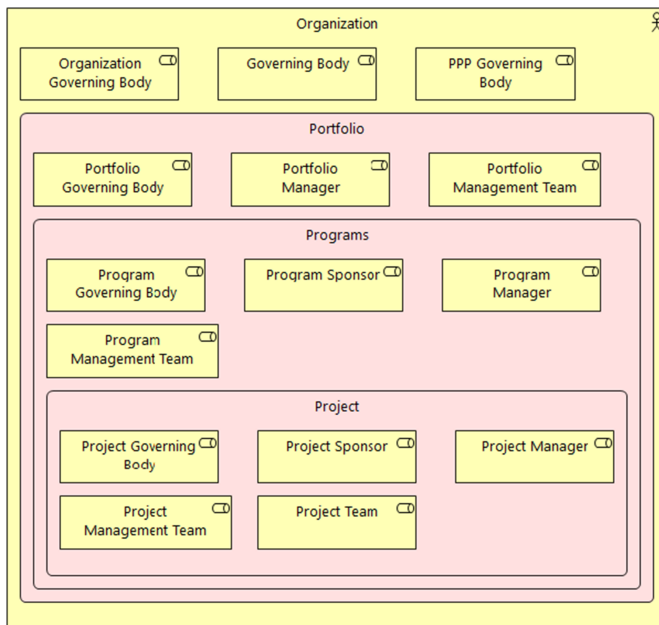


Fig. 4. ArchiMate representation of the framework of the roles associated with PPP governance within an organization [5].

As mentioned in [10], the roles present in ISO’s 21500 [15], 21503 [16], 21504 [17] and 21505 [17], are used as a reference in other methodologies. These roles are presented in Sect. 2.2 for PPP and organizational governance and Sect. 2.3 for project, program and portfolio management, and are used in the reference architecture.

3.3 Identify Role Concerns

To identify the concerns of each role, the organization documentation is analyzed (where the roles are identified) and a table is filled, where it is possible to check which concerns and responsibilities associated with each role.

This approach provides an easy access to the concerns of each role, which is useful for the next steps (Fig. 5).

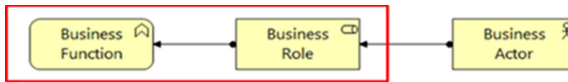


Fig. 5. View of role matching with their concerns identified in the ArchiMate Business Function viewpoint [6].

3.4 Associate Roles with Layers

To organize roles in the respective layers, it is necessary to have access to the concerns/responsibilities of each role as well as the definition of each layer of the architecture.

The mapping is performed layer by layer:

- **Business Governing Layer:** Determines the vision and strategy for the entire organization. It consists of one or more management committees operating at director level. This is where priorities are set, investment decisions are made, and resources allocated. In ISO 21505 [18], the governing body role is introduced as “a person, group or entity responsible for the governance of the organization or part of the organization” which is then associated with this layer.
- **Steering Layer:** Provides guidance for the project, program or portfolio, keeping each one of them focused on the goals. PPP Governing Body is “the person, group or entity responsible for the governance of projects, programs and portfolios as an organization, supported by a set of interrelated organizational processes by which an organization prioritizes, selects and allocates resources to achieve organizational objectives” [18]. An organization may have a specific governing body for the project ecosystem, the program ecosystem and the portfolio ecosystem.
- **Directing Layer:** Sponsors the project, program or portfolio. It mobilizes the necessary resources and monitors the performance of the project, program or portfolio to achieve its objectives. Project Sponsor is defined as “the person who authorizes the project, makes executive decisions and solves problems above the authority of the project manager” [15]. The Program Sponsor is defined as the person “responsible for the overall program strategy and its implementation defence” [16].
- **Managing Layer:** is focus on the daily operations of the project, program or portfolio, organizing, monitoring and controlling the work to produce the intended deliverables and implement them in the organization. Managing Layer members report to

Directing Layer. In this layer, the project manager is the person who “leads and manages the project activities and is responsible for its completion” [15], supported by the project management team. The program manager is the person “responsible for project performance, program and coordinating its components” [16], supported by the program management team. Finally the portfolio manager is the person “responsible for applying the portfolio management supported by the portfolio management team” [17].

- Performing Layer: responsible for the work of the project, program or portfolio, producing the deliverables and implementing them in the organization. Performing Layer members report to the Managing Layer. Associated with this layer is the Project Team role that “carries out the project activities” [15] (Fig. 6).

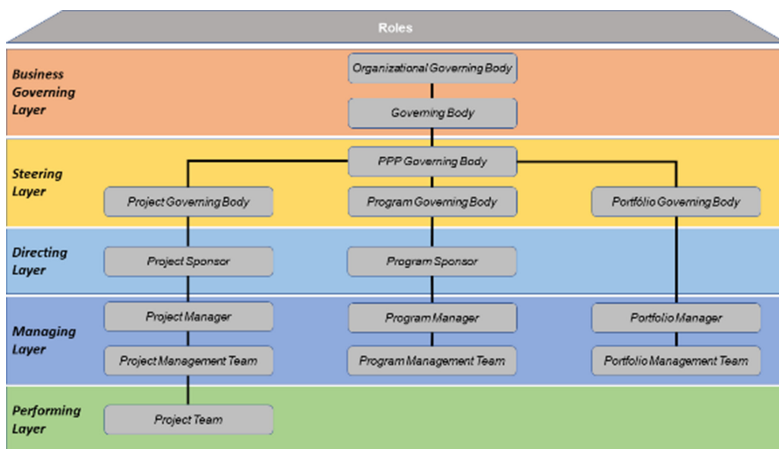


Fig. 6. Mapping the roles associated with PPP Governance in the respective layers [6].

3.5 Associate Competences with Roles

To associate which competencies are required for each role, it is essential to consider the competency indicators present in IPMA ICB, as well as the concerns of each of them.

Through a comparative analysis between the concerns of each role as well as the indicators of each competence, a mapping between the roles and the competences is performed (Appendix A).

3.6 Associate Competences with Layers

Through the mapping of roles in layers and the association of competencies with roles, a mapping of layered competencies.

For each layer of the architecture, the roles that belong are identified, after that the competencies of these same roles are also identified and finally the competencies are associated with these same layers.

3.7 Model in ArchiMate by Layer the Roles and Their Competencies

Through the mapping of roles in layers and the association of competencies with roles, a mapping of layered competencies. For each layer of the architecture, the roles that belong are identified, after that the competencies of these same roles are also identified and finally the competencies are associated with these same layers.

To correctly model in ArchiMate, it is necessary relate the terms “role” and “competence” with the elements present in ArchiMate, namely the elements “business role” and “capability” – see Table 3 (Fig. 7).

Table 3. Mapeamento entre termos da arquitetura de referência e elementos do ArchiMate [6].

Reference architecture concept	Element in ArchiMate
“Role” – “Function or something that someone has” [28])	“Business Role” – “Responsibility to perform a specific behavior” [32]
“Competence” – “applying knowledge, skills and abilities to achieve desired results” [14]	“Capability” - “represents a resource (knowledge, skills, abilities) that an element of the active structure, such as an organization, person or system has or should have” [32]



Fig. 7. Viewpoint of the mapping in ArchiMate of the roles and respective competences [6].

4 Demonstration

The demonstration of the use of the PPP governance methodology is performed in a Government owned company which for confidentiality reasons is named DemoCorp. DemoCorp is an organization with 740 employees, that has in the PMO (Project Management Office) the governance model that controls an ecosystem with portfolios, programs and projects, promoting organizational alignment and increasing the levels of efficiency and effectiveness in project development. DemoCorp is rated at maturity level 4 according to Gartner’s IT Score for Program & Portfolio Management maturity model.

The demonstration is carried out step by step, through a process similar to the one described in the “Solution Proposal” section. The DemoCorp methodology is represented in the layers of the reference architecture supporting a comparative analysis between the methodology of DemoCorp and the PPP governance model used in the reference architecture.

With the demonstration one may conclude that the number of roles is less in the DemoCorp model compared to the reference architecture governance model. Upon further analysis the following findings are identified:

- In the business governing layer, there is one more role in the DemoCorp model than in the reference architecture's PPP governance model.
- There are also fewer roles in the project ecosystem than in the DemoCorp model. In the Managing Layer there is no project management team that assists the project manager which can cause work overload on the project manager. Steering Layer also doesn't have a Project Governing Body, with Project Management Officer's role being responsible for its functions.
- However, a project sponsor is added to realize the benefits of the project in the organization linking the project and its implementation in the organization.
- In both the program and portfolio ecosystems, a role in the Managing Layer is removed and there is no program management team or portfolio management team, which means that both program and portfolio managers may be overloaded. In these ecosystems there is no program governing body or portfolio governing body. Similar to the project ecosystem, its functions are performed by the project management officer (Fig. 8).

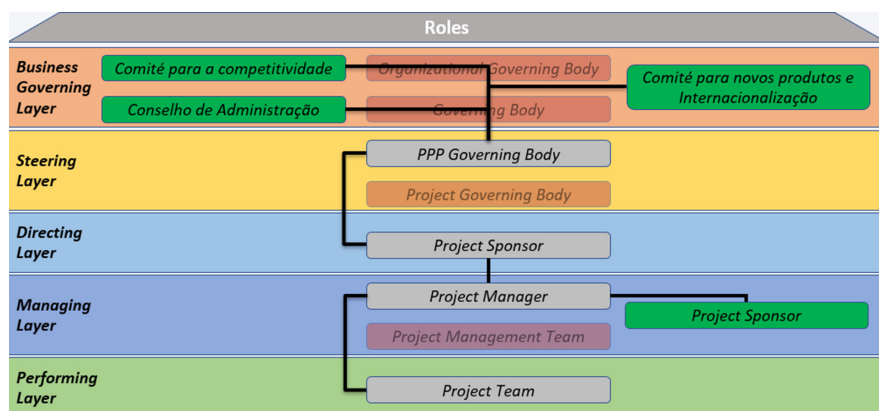


Fig. 8. Analysis of role deviations [5].

At the level of missing competencies or competencies added through the demonstration one can conclude that the number of competences is less in DemoCorp model compared to the reference architecture governance model (respectively 91 and 122 competences), see (Appendix B).

Performing a further analysis, one found that in the set of both governance models, 156 competences were identified, of which 64 are in accordance with both models, 27 are introduced by DemoCorp governance model and 65 only appear in the governance model of the reference architecture (Fig. 9).

Through the demonstration it is possible to verify that DemoCorp PPP governance methodology has some gaps, both in terms of roles and competencies associated with it. In terms of roles, there is an absence of teams that help portfolio managers, programs and projects, as well as a specific governing body for both projects, programs and portfolios.

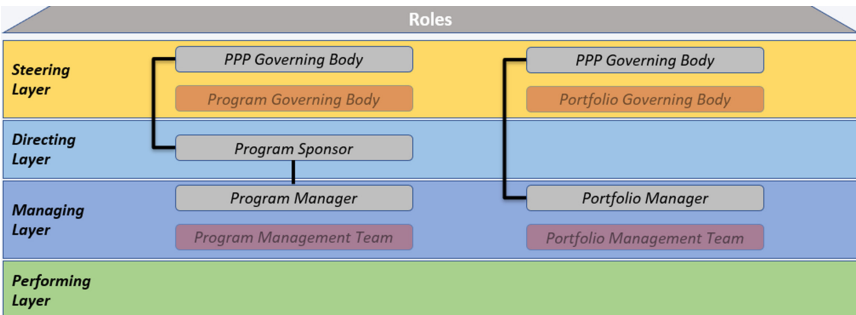


Fig. 9. Analysis of role deviations (program and portfolio ecosystem) [5].

Regarding competencies, it is noted that there are 65 competencies not considered in DemoCorp methodology, suggesting a revision of the methodology processes in order to decrease this figure.

It is important to notice that the addition of roles and the 27 skills that are only included in the DemoCorp model can be justified by its context and needs.

5 Evaluation

The evaluation of this research is performed considering DemoCorp demonstration. The evaluation is performed in three steps. In Sect. 5.1 the artefacts that support the solution, namely the reference architecture, are evaluated. In Sect. 5.2 the evaluation of the solution objectives is performed.

In Sect. 5.3 the proposed solution is compared with existing ones.

5.1 Reference Architecture

To assess the reference architecture, PPP management professionals were asked to perform a similar process to steps 3.4 - Associate Roles with Layers and 3.5 - Associate Competencies with Roles.

These two steps were chosen because they are the structural steps in the creation of the reference architecture.

The result of step 3.4 coincided 100% with the reference architecture and with the step 3.5 we concluded that of the total of 317 competences addressed by the reference architecture and by the respondent or both, 57.4% comply, i.e. according to both (reference and respondent architecture), 6.6% are only addressed by the reference architecture and 36% are only addressed by the respondent.

The fact that only a low percentage (6.6%) of competences is not addressed by the respondents is a good indicator for validating the association of skills with roles in the reference architecture (because it shows that most of the competences present in the reference architecture were identified by the respondents).

5.2 Solutions Benefits

The proposed solution provides a reference architecture that can serve as a basis for comparing PPP governance models. This reference architecture allows to verify the failing elements that may be influencing the functioning of the PPP governance model and also to find deviations that may be related to the context of the organization that holds the PPP governance model to be compared.

It facilitates the identification/understanding of failing elements in PPP governance models in a classical approach to project management. Through the demonstration in Sect. 4 we find that it is possible to perform an analysis of both competencies and roles associated with the PPP governance model.

We can see which roles and competencies have been added or are flawed in DemoCorp methodology based on the reference architecture presented in Sect. 3.

It highlights additional roles and relationships by comparison, allowing to explore role patterns or redundancy, better understanding the value offer of one methodology over another. By adding and removing roles (that the method allows to identify), patterns may be created considering the different contexts of project execution. An example of this is the addition of the project sponsor, and associated skills, working with the project manager to ensure that all project benefits are absorbed by the organization. This addition can be considered a necessary standard when dealing with a certain type of projects.

Another benefit of the proposed solution is on the verification on how the skills required for proper PPP governance are addressed when roles are added or removed. ArchiMate modelling makes it easier to analyse competency handling when roles are added or removed.

Finally, the proposed solution allows access to a reality that comprises only one project, or one project ecosystem and its associated methodologies, including programs and portfolios. The method allows comparing PPP governance models that encompass both a project-only ecosystem, an ecosystem with programs and projects, and an ecosystem with portfolios, programs and projects.

5.3 Related Work Comparison

5.3.1 Roles

In contrast to the studied PPP governance models, namely the APMBok and PM2 models, which address an ecosystem where there are only projects, the PPP governance model used in the reference architecture, similar to the PMBOK model, besides supporting a project ecosystem, it also supports a program ecosystem and a portfolio ecosystem.

5.3.2 Competences

IPMA ICB, which is the main source for the competencies used in the reference architecture, presents the cross-referencing of the competencies presented with ISO 21500 (from which the roles and their concerns/responsibilities associated with project management were taken) and with ISO 21504 (from which the roles and responsibilities/concerns associated with portfolio management were taken).

After analysing the result of this intersection with what is presented in the reference architecture, we find that with respect to the ISO 21500 the identified roles don't perform two competences ("Selfreflection and Self-management" and "negotiation") which are addressed in the intersection of ISO with IPMA ICB. With respect to ISO 21504, the roles identified therein perform all the competences addressed at the intersection of this ISO with the IPMA ICB. The reference architecture by also mapping the competencies of IPMA ICB with the roles taken from ISO 21503 and ISO 21505 also makes a possible intersection of these ISOs with IPMA ICB.

5.3.3 Architecture Levels

None of the studied researches present simultaneously the two levels present in the reference architecture or present only the level with the essential roles for the correct governance of PPP or present only the level with the competences.

Therefore, this architecture is innovative because it allows the roles to be associated with the skills they must perform.

5.3.4 Other Comparing

The reference architecture proposed may also be compared with maturity models since they the evaluation of PPP governance methodologies by assigning them a level of maturity.

All maturity models are concerned with PPP governance processes and with the creation the creation of a structure, that allows for this same governance. None of them deepens this same structure, namely the roles it should contain and the skills they must perform.

Therefore this research allows to compare the PPP governance structures considered essential in the maturity models, at a level of roles and competencies that it should contain.

6 Conclusions

Projects are important for organizations to reach their strategy and keep up with the competitive and changing market.

For projects to be successfully delivered, complying with estimated time, cost and resources, it is necessary that correct project management is followed, in an ecosystem with projects, programs and portfolios.

All the roles and responsibilities associated with correct PPP management must be established by the PPP governance model, which must follow the principles and values of the organizational governance model.

This research presents a reference architecture, this architecture makes possible to analyze two or more PPP governance models and find differences between them. For that, this research allows to check differences between models and choose the model that best suit with each organization scope.

The reference architecture consists in a layered architecture, when in each layer we can find the respective roles identified in ISO's 21500, 21503, 21504 and 21505. These

documents are quite referred in so much other project management methodologies. In the reference architecture we can find too the competences related to project management divided in the architecture layers. This research adopts an Enterprise Architecture approach for modeling roles and the respective competences in the different ecosystems (portfolios, programs or projects), using ArchiMate modeling language.

For prove the useless of the reference architecture created, we use a DemoCorp and its governance model. In fact, when analyze the different governance models (Reference Architecture Governance Model VS DemoCorp Governance Model) we find differences between roles and competences.

6.1 Contributions

This research contributes to the scientific community in the different ways. The major contribution is the reference architecture created. This architecture is the first architecture that through the relation between the roles presented in ISO's and the competences put the competences organized in the project management layers. The layered architecture structure is another innovation that allows a better comprehension of the differences between PPP governance models and can help to choose the best model for each company, due its context.

While we work on the main objective that was a PPP reference architecture we contribute with other things, including:

- Highlighting new roles, new competences and exploring patterns or redundancy, better understanding the advantages among governance models.
- Crossing between IPMA ICB with ISO 21503 and ISO 21505. When analyzing the IPMA ICB, it was found that the Appendix resents the crossing of the competences presented with ISO 21500, from where the roles and their concerns/responsibilities associated with project management were taken and with ISO 21504 from where the roles and their concerns/responsibilities in relation to portfolio management. The fact that the reference architecture presents the mapping of the competences of the IPMA ICB with the roles taken from ISO 21503 and ISO 21505 makes a possible crossing of these ISOs with the IPMA ICB that had not yet been carried out.
- The capacity of verifying how the competences for proper PPP governance is addressed when roles are added or removed.

6.2 Limitations

During our research several limitations were identified, presenting new research opportunities to improve the reference architecture.

The limitations found are as follows:

- I. the fact that one role may have much competences and responsibilities and may be associated with different layers;
- II. when we map competencies to roles, through the IPMA ICB, we can find a set of indicators for the same competence. Considering it, the same role may be associated with different competences taking into account differ-ent indicators;

- III. the attribution of competencies to roles, in most cases, is not something that is straightforward after analyzing concerns and responsibilities, bringing some degree of uncertainty to the approach proposed.

6.3 Future Work

This research and its limitations can allow some future work opportunities. This reference architecture can be used in different contexts with the goal of find and establish different patters of roles and competencies needed for each context. Due to the limitation of the mapping between roles and competences, a deeper investigation must be taken to link roles with competences in a stronger way.

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Appendix A

See Table 4.

Table 4. Mapping the competences associated with each role taking into account the indicators.

Role	Competence	Indicator
Global		
Organization Governing Body	Result Orientation	1, 2, 3
	Organisation and information	2, 4, 5
Governing Body	Strategy	2, 3
	Governance, structures and processes	1, 2, 3, 4
	Compliance, standards and regulations	1, 2, 3, 4, 5, 6
	Culture and values	1, 2, 3
	Personal integrity and reliability	3
	Leadership	1, 3, 5
	Teamwork	4
	Conflict and crisis	3
	Relationships and engagement	1
	Result Orientation	3
	Requirements, Objectives and Benefits	5, 6
	Scope	6
	Organisation and information	3, 4, 5
	Plan and Control	1, 5, 7
	Stakeholders	1, 2, 3, 4, 5
PPP Governing Body	Strategy	1
	Culture and values	1, 2, 3
	Compliance, standards and regulations	1, 2, 3, 4
	Governance, Structures and Processes	1, 2, 3, 4
	Result Orientation	2
	Leadership	4
	Resourcefulness	5
	Personal integrity and reliability	1, 2, 3, 4, 5
	Personal Communication	1, 2, 3
	Finance	7
	Resources	2, 3, 4, 6
	Requirements, Objectives and Benefits	3, 7
	Scope	2, 4
	Plan and Control	7
	Organisation and information	1, 2, 3, 4, 5
	Quality	1
	Procurement	1, 2
	Risk and opportunities	1
	Stakeholders	1, 2, 3, 4, 5

(continued)

Table 4. (continued)

Role	Competence	Indicator
Projects		
<i>Project Manager</i>	Resul orientation	1, 4, 5
	Personal integrity and reliability	3, 4, 5, 1
	Personal Communication	1, 2, 3
	Leadership	1, 2, 3, 4, 5
	Teamwork	1, 2, 3, 4
	Conflict and crisis	1, 4
	Resourcefulness	4, 5
	Relationships and engagement	1, 2
	Governance, structures and processes	1
	Compliance, standards and regulations	1, 2, 3, 4, 5
	Power and interest	1, 2, 3
	Culture and values	1, 2, 3
	Requirements, Objectives and Benefits	1, 3, 5, 7
	Design	2, 3, 6, 7
	Scope	1, 3, 4, 5, 6
	Time	2, 3, 4, 5, 6
	Quality	1, 2, 5
	Finance	4
	Organisation and information	2
	Plan and Control	2, 7, 9, 10, 11, 12,13
<i>Project Manager</i>	Resources	1, 2, 3, 4, 5, 6
	Stakeholders	2
	Procurement	5
	Risk and opportunities	1, 2, 3, 4, 5
<i>Project Management Team</i>	Resul orientation	1, 4, 5
	Personal integrity and reliability	1, 3, 4, 5
	Personal Communication	1, 2, 3
	Leadership	1, 2, 3, 4, 5
	Teamwork	1, 2, 3, 4
	Conflict and crisis	1, 4
	Resourcefulness	4, 5
	Relationships and engagement	1, 2
	Governance, structures and processes	1, 4
	Compliance, standards and regulations	5
	Power and interest	1, 2, 3
	Culture and values	1, 2, 3
<i>Project Management Team</i>	Requirements, Objectives and Benefits	1, 3, 5, 7
	Design	2, 3, 6, 7
	Scope	1, 3, 4, 5, 6
	Time	2, 3, 4, 5, 6

(continued)

Table 4. (continued)

Role	Competence	Indicator
Projects		
Project Management Team	Quality	1, 2, 5
	Finance	[4]
	Plan and Control	2, 7, 9, 10, 11, 12,13
	Resources	1, 2, 3, 4, 5, 6
	Stakeholders	2
	Procurement	5
	Risk and opportunities	1, 2, 3, 4, 5
Project Team	Personal integrity and reliability	5
	Risk and opportunities	1, 2, 3, 4, 5
	Plan and Control	11, 12, 13
Project Sponsor	Result Orientation	1, 5
	Personal integrity and reliability	1, 3, 4
	Personal Communication	1, 3
	Relationships and engagement	1
	Leadership	5
	Conflict and crisis	1, 2, 3, 4
	Governance, structures and processes	4
	Strategy	2, 3
	Scope	3
	Requirements, Objectives and Benefits	2, 4, 6
Project Sponsor	Plan and Control	11, 12, 13
	Risk and opportunities	1, 2, 3, 4, 5
	Organisation and information	2
Project Governing Body	Strategy	1, 3
	Governance, structures and processes	4, 5
	Compliance, standards and regulations	1, 2, 3, 4, 5
	Relationships and engagement	2
	Leadership	1, 2, 3, 4, 5
	Requirements, Objectives and Benefits	5
	Organisation and information	2, 4, 5
	Quality	1, 2, 5, 6
	Resources	4
	Procurement	2
	Plan and Control	9
	Risk and opportunities	1, 2, 3, 4, 5
	Stakeholders	2
	Change and transformation	5

(continued)

Table 4. (continued)

Role	Competence	Indicator
Programs		
Program Sponsor	Strategy	1, 3
	Personal integrity and reliability	4
	Personal Communication	1, 2
	Relationships and engagement	5
	Leadership	5
	Teamwork	4
	Conflict and crisis	1, 2, 3, 4
	Result Orientation	1
	Design	5
	Requirements, Objectives and Benefits	5
	Organisation and information	2, 3
	Plan and Control	2, 11
	Stakeholders	1, 2, 3, 4, 5
Program Manager	Governance, structures and processes	2
	Power and interest	1, 2, 3
	Personal integrity and reliability	1, 3, 4
	Personal Communication	1, 2
	Relationships and engagement	3, 4
	Leadership	1, 3, 4, 5
	Teamwork	3, 4
	Conflict and crisis	1, 2, 3, 4
	Negotiation	5
Program Manager	Result Orientation	1
	Design	6, 7, 8
	Requirements, Objectives and Benefits	1, 2, 4, 5
	Organisation and information	1, 4
	Resources	2, 4, 5, 6
	Plan and Control	2, 5, 7, 8
	Finance	2
	Stakeholders	1, 2, 3, 4, 5
Program Management Team	Select and balance	1, 5, 7
	Governance, structures and processes	1, 2, 4
	Design	6, 7
Program Governing Body	Requirements, Objectives and Benefits	3
	Strategy	1, 3
	Governance, structures and processes	4, 5, 6, 7
	Compliance, standards and regulations	1, 2, 3, 4, 5, 6
	Culture and values	2, 3
	Personal integrity and reliability	1, 3, 4
	Personal Communication	1

(continued)

Table 4. (continued)

Role	Competence	Indicator
Programs		
Program Governing Body	Leadership	1, 2, 3, 4, 5
	Teamwork	4
	Result Orientation	1
	Design	4, 5
	Metas Objetivos e Benefícios	1, 5
	Scope	4
	Time	7
	Organisation and information	2
	Quality	3
	Finance	3, 4
	Resources	1, 5
	Plan and Control	1, 2, 5, 11
	Risk and opportunities	1, 4
	Change and transformation	1, 2, 4, 5, 6
	Select and balance	1, 2
Portfolios		
Portfolio Manager	Strategy	1, 2, 4, 5
	Governance, structures and processes	3, 5, 6, 7
	Compliance, standards and regulations	1, 2, 3, 4, 5
	Culture and values	2
	Leadership	5
	Conflict and crisis	2
Portfolio Manager	Negotiation	1, 4, 5
	Result Orientation	1
	Design	1, 7
	Requirements, Objectives and Benefits	1, 2, 3, 5
	Time	1, 4, 5, 6
	Organisation and information	1
	Resources	1, 3, 5
	Plan and Control	1, 3, 4, 5, 6, 8, 11
	Risk and opportunities	1, 4, 5
	Stakeholders	1, 3
	Change and transformation	1, 4, 5, 6
	Select and balance	1, 3, 5, 6, 7
Portfolio Management Team	Governance, structures and processes	4
Portfolio Governing Body	Strategy	1, 3
	Governance, structures and processes	3, 5, 6, 7
	Compliance, standards and regulations	1, 2, 3, 4, 5, 6
	Culture and values	2

(continued)

Table 4. (continued)

Role	Competence	Indicator
Portfolios		
<i>Portfolio Governing Body</i>	Personal integrity and reliability	1, 3, 4
	Personal Communication	1
	Leadership	1, 2, 3, 4, 5
	Teamwork	5
	Negotiation	4
	Result Orientation	1
	Design	4
	Requirements, Objectives and Benefits	1
	Organisation and information	2
	Quality	3
	Finance	8
	Resources	4, 5, 6
	<i>Procurement</i>	4
	Plan and Control	7
	Risk and opportunities	1, 4
	Select and balance	1, 2, 6, 7

Appendix B

See Table 5.

Table 5. Differences between the reference architecture and the model to be compared at the level of competencies by layer.

Competences/Layer		
Reference Architecture	Competence	Democorp Architecture
Global		
Business Governing Layer		
X	Strategy	X
X	Governance, structures and processes	X
X	Compliance, standards and regulations	
X	Culture and values	
X	Personal integrity and reliability	
	Personal Communication	X
X	Relationships and engagement	
X	Leadership	X
X	Teamwork	
X	Conflict and Crisis	
X	Result orientation	
X	Requirements, objectives and benefits	X
X	Scope	
X	Organisation and information	X
	Resources	X
X	Plan and control	X
X	Stakeholders	
	Select and balance	X

(continued)

Table 5. (continued)

Competences/Layer		
Reference Architecture	Competence	Democorp Architecture
Global		
Steering Layer		
X	Strategy	X
X	Governance, structures and processes	
X	Compliance, standards and regulations	
X	Culture and values	X
X	Personal integrity and reliability	X
X	Personal Communication	X
	Relationships and engogement	X
X	Negotiation	
X	Leadership	X
X	Teamwork	
X	Resourcefulness	
X	Result orientation	X
X	Requirements, objectives and benefits	
X	Scope	X
X	Resources	
X	Time	X
X	Organisation and information	X
X	Quality	
X	Finance	
X	Design	
X	Resources	X
X	<i>Procurement</i>	
X	Plan and control	
X	Risk and opportunitieess	
X	<i>Stakeholders</i>	
X	Change and transformation	
X	Select and balance	
Projetos		
Directing Layer		
X	Strategy	
X	Governance, structures and processes	X
	Compliance, standards and regulations	X
	Power and interest	X
	Culture and values	X
X	Personal integrity and reliability	X
X	Personal Communication	X
X	Relationships and engogement	X
X	Leadership	X
	Teamwork	X
X	Conflict and Crisis	X
X	Result orientation	
	Design	X
X	Requirements, objectives and benefits	X
X	Scope	
X	Organisation and information	X
	Finance	X
	Resources	X
	<i>Procurement</i>	X

(continued)

Table 5. (continued)

Competences/Layer		
Reference Architecture	Competence	Democorp Architecture
Projetos		
Directing Layer		
X	Plan and control	X
X	Risk and opportunitiess	
	Stakeholders	X
	Change and transformation	X
Managing Layer		
	Strategy	X
X	Governance, structures and processes	X
X	Compliance, standards and regulations	X
X	Power and interest	X
X	Culture and values	
X	Personal integrity and reliability	X
X	Personal Communication	X
X	Relationships and engagemet	X
X	Leadership	X
X	Teamwork	X
X	Conflict and Crisis	X
X	Resourcefulness	
X	Result orientation	
X	Design	X
X	Requirements, objectives and benefits	X
X	Scope	X
X	Time	
	Organisation and information	X
X	Quality	X
X	Finance	X
X	Resources	X
X	Procurement	
X	Plan and control	X
X	Risk and opportunitiess	X
Managing Layer		
X	Stakeholders	
	Change and transformation	X
Performing Layer		
X	Personal integrity and reliability	X
X	Plan and control	X
X	Risk and opportunitiess	
Programas		
Directing Layer		
X	Strategy	
	Governance, structures and processes	X
X	Personal integrity and reliability	X
X	Personal Communication	X
X	Relationships and engagemet	X
X	Leadership	
X	Teamwork	
X	Conflict and Crisis	
X	Result orientation	
X	Design	X
X	Requirements, objectives and benefits	

(continued)

Table 5. (continued)

Competences/Layer		
Reference Architecture	Competence	Democorp Architecture
Programas		
Directing Layer		
	Time	X
X	Organisation and information	X
X	Plan and control	X
X	Stakeholders	
Managing Layer		
X	Governance, structures and processes	X
X	Power and interest	
X	Personal integrity and reliability	X
X	Personal Communication	X
X	Relationships and engogement	X
X	Leadership	
X	Teamwork	X
X	Conflict and Crisis	
X	Negotiation	
X	Result orientation	
X	Design	
X	Requirements, objectives and benefits	X
	Scope	X
	Time	X
X	Organisation and information	X
X	Finance	
X	Resources	
X	Plan and control	X
X	Stakeholders	
X	Select and balance	
X	Strategy	
X	Governance, structures and processes	X
X	Compliance, standards and regulations	
X	Culture and values	
	Personal integrity and reliability	X
	Personal Communication	X
	Relationships and engogement	X
X	Leadership	
	Teamwork	X
X	Conflict and Crisis	
X	Negotiation	
X	Result orientation	
X	Design	
X	Requirements, objectives and benefits	X
	Scope	X
X	Time	X
X	Organisation and information	X
	Finance	X
X	Resources	X
X	Plan and control	X
X	Risk and opportunitiess	
X	Stakeholders	
X	Change and transformation	
X	Select and balance	

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